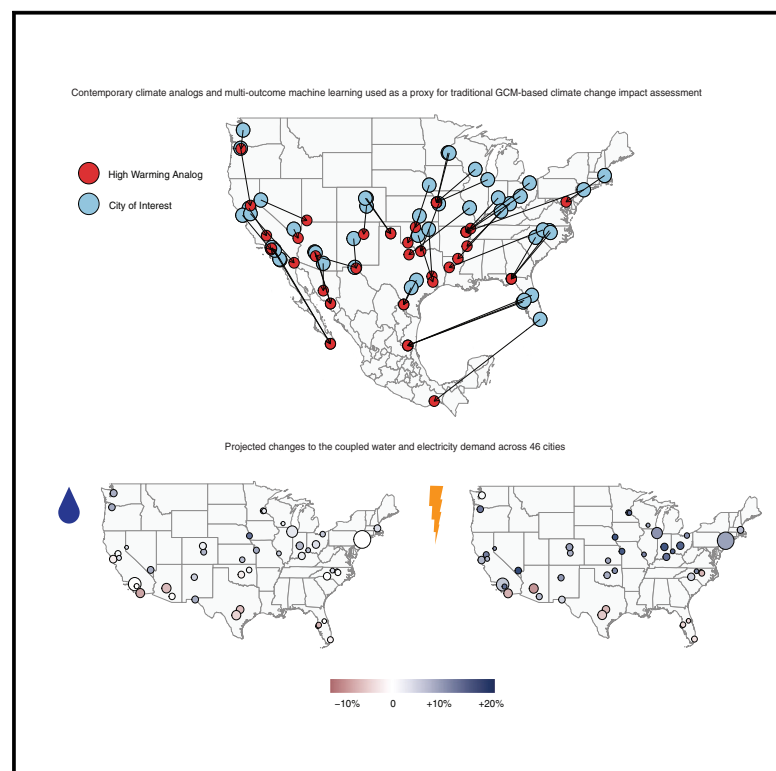


Contemporary climate analogs project strong regional differences in the future water and electricity demand across US cities

Graphical abstract



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In brief

Understanding the impacts of climate change on water and electricity systems is crucial for building resilience and working to ensure an adequate supply of critical services. By integrating contemporary climate analogs with state-of-the-art machine learning, this study projects climate-induced shifts in water and electricity demand of up to 15% and 20% in major US cities, with strong regional differences across the country.

Highlights

- We integrated climate analogs and machine learning for climate impact assessment
- Climate-induced shifts in water and electricity use were explored for 46 US cities
- Strong regional differences in the projected demand are shown across US
- Water and electricity demand projected to increase up to 15% and 20% in some cities



Article

Contemporary climate analogs project strong regional differences in the future water and electricity demand across US cities

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<https://doi.org/10.1016/j.oneear.2023.10.004>

SCIENCE FOR SOCIETY Climate change is impacting how society uses water and electricity, from increased air conditioning use to greater use of residential irrigation for lawns and landscaping. Understanding how these impacts will continue to evolve is critical for the management of our infrastructure systems, but there is a lack of available models to do such impact assessments. In this study, we use an integration of climate analogs and machine learning to conduct a computationally efficient climate change impact assessment for residential water and electricity use across 46 cities in the US. Our results depict strong regional shifts in the projected water and electricity demand because of future climate change should we fail to mitigate the crisis.

SUMMARY

Water and electricity systems are highly interconnected in terms of supply (e.g., electricity is used in the treatment and distribution of water) and demand (e.g., appliances that use water within homes also use electricity). Understanding the impact that climate change may have on coupled water and electricity demand is a crucial task for ensuring resilient planning and sustainable management of water and energy infrastructure. Here, we leverage an efficient methodology that integrates contemporary climate analogs and machine learning to project the city-level coupled water and electricity demand of 46 major US cities into the future. The results show strong regional differences, with some cities possibly experiencing increases in summer water and electricity demand of up to 15% and 20%, respectively, because of climate change. Obtaining projections of future water and electricity demand under climate change is critical for building resilient infrastructure in the face of climate change.

INTRODUCTION

Globally, cities are growing at an increasingly fast rate, with nearly 70% of the world's population estimated to live in cities by 2050.¹ As urban populations grow, so will the need for critical services such as water and electricity. In fact, the provisioning of these services is of paramount importance for sustainable development.² The compound effects of population growth and anthropogenic climate change will constrain the supply of water and electricity while also increasing the demand for these resources, straining the existing water and energy infrastructure systems. For example, several studies have demonstrated the possible impacts of climate

change on water supply across the globe, from the state of California³ to the country of Iran.⁴ Likewise, a number of studies have shown similar impacts on the energy system, including energy supply⁵ and demand.^{6,7} More recently, research has shifted toward investigating impacts on interconnected infrastructure, such as the water-electricity demand nexus.⁸ In general, water-energy nexus studies primarily focus on the supply side, such as the nexus between water supply and electricity generation through hydropower^{9,10} or the use of water by electricity generators for cooling purposes.^{11–13} There is, however, a growing body of work that investigates the demand side of the nexus, looking in particular at the simultaneous consumption of water and electricity.^{14–16}

Within the literature on the urban water-energy nexus (see Newell et al.¹⁷ and Newell and Ramaswami¹⁸ for comprehensive reviews of the subject), there has been some work across the US that has focused on the energy used by water treatment plants¹⁹ as well as the water used for electricity generation.²⁰ However, in terms of the water-electricity demand nexus, the focus tends to be on smaller scales, such as a single city (e.g., Dale et al.²¹ and Escriva-Bou et al.¹⁵ focused on municipalities in California—Sacramento and the East Bay area, respectively) or a small group of cities (e.g., Venkatesh et al.²² studied four different cities across Europe and Canada, while Obringer et al.^{8,16} limited their studies to six Midwestern cities). Given that management of water and electricity systems occurs at multiple scales (e.g., ranging from local utilities to federal management entities), there is a need to expand urban water-energy nexus analyses to consider larger scales.

Additionally, studies that leverage data-driven modeling of climate change impacts on the *climate-sensitive* portion of end-use demand (i.e., the portion of the total consumption that is influenced by climatic conditions, such as outdoor landscaping or air conditioning use) primarily involve first characterizing the relationship between climate and demand in the observational space and then projecting those relationships into the future using general circulation model (GCM)-derived climate data. For example, Maia-Silva et al.²³ and Obringer et al.²⁴ leveraged GCM data to project the climate change impacts on electricity demand, while Obringer et al.⁸ utilized similar data for a water-energy nexus study. While this practice is beneficial and can provide accurate representations of future demand, the incorporation of GCM-based data is computationally expensive, rendering it unsuitable for local utilities without access to high-performance computing and necessary expertise. In this regard, finding alternatives that have lower computational costs is a practical need for building resilient water and electricity infrastructure.

Overall, it is critical that urban water-energy nexus studies account for anthropogenic climate change, which is one of the most pressing challenges society faces. To do this, there is a need to create transferable frameworks that can be readily applied to large-scale analyses. Additionally, it is beneficial, particularly from a practitioner's perspective, to evaluate quick but reliable techniques for obtaining future climate data that are more user friendly and less computationally intensive. Here, we use one such approach—climate analogs. The analog approach has been adopted to study a wide variety of locations, including cities in the US,²⁵ Africa,²⁶ and China,^{27,28} as well as for communication and policy studies.^{29,30} Climate analogs are determined through statistical processes that identify the location where the *current* climate is analogous to the *future* climate in another location.

This study aims to fill a significant gap in the literature by presenting a computationally inexpensive projection of the climate-related changes in future water and electricity demand across a number of major US cities while also emphasizing the importance of considering the water-electricity nexus in such analyses. Additionally, the use of climate analogs as the primary source of future climate data provides an opportunity to communicate climate change impacts to US citizens and decision-makers in a more intuitive way than traditional GCM-based studies, which often do not have a comparative analysis basis.

In this study, we leverage climate analogs²⁵ to characterize the future climate in 46 major cities (population > 250,000) across the US. By integrating contemporary climate analogs with state-of-the-art machine learning (i.e., multivariate tree boosting³¹), we use the observed climate data in the analogs to project the coupled residential water and electricity demand for each city of interest. Ultimately, this leads to a more computationally efficient and accessible means of conducting a climate impact assessment within the water and electricity sectors. Using the high emissions scenario (RCP8.5), our results indicate strong regional differences across the US, with some cities projected to experience an increase of up to 15% and 20% in water and electricity demand, respectively. The aim of this study was to present a novel technique for conducting a climate change impact assessment while also exploring possible changes to the water-electricity demand in the future. To that end, we provided a computationally efficient means of understanding future climate impacts. Moreover, we discuss the results of model in terms of the projected changes for water-electricity demand as well as the implications for the planning and management of the larger infrastructure systems.

RESULTS

Methods summary

The methodology of this work was composed of two modeling steps: (1) using observational data for model training and (2) using climate analog data to investigate the impact of future climate change. Throughout the remainder of this paper, we will refer to the first step as a “predictive model” and to the second step as a “projection model” to differentiate between the two steps. The model was built using observational data from the North American Regional Reanalysis (see Experimental procedures and Figure 1B for input variables). In this study, we analyzed the data for three main seasonal blocks: summer (June–September), winter (December–March), and intermediate (April, May, October, and November). Throughout the majority of this study, however, we focus on the summer months because the consumption during these months accounts for 30%–50% of the total annual consumption (Figure 1A). Moreover, the summer consumption has been shown to be more sensitive to climatic conditions than the consumption during the remainder of the year.^{8,32} To this end, it is important to note that all references to the water and electricity consumption values refer to the summer months unless stated otherwise.

To measure model performance of the prediction model, we calculated R^2 and normalized root-mean-square error (NRMSE). In terms of R^2 , the median value for the climate-driven portion of electricity and water consumption was 0.70 and 0.56, respectively, suggesting that the model does explain a notable percentage of variance for the majority of the cities. However, the values did range from 0.36–0.90 for electricity consumption and 0.31–0.84 for water consumption, indicating the varying degrees of climate sensitivity of demand across the US. In other words, it indicates that there are some cities for which the water and electricity demand is not completely explained by the climate variables. Figure 1C shows that, in terms of water consumption, some cities in the southeast as well as the southwest do not have as close of a fit as other areas. However, similar work

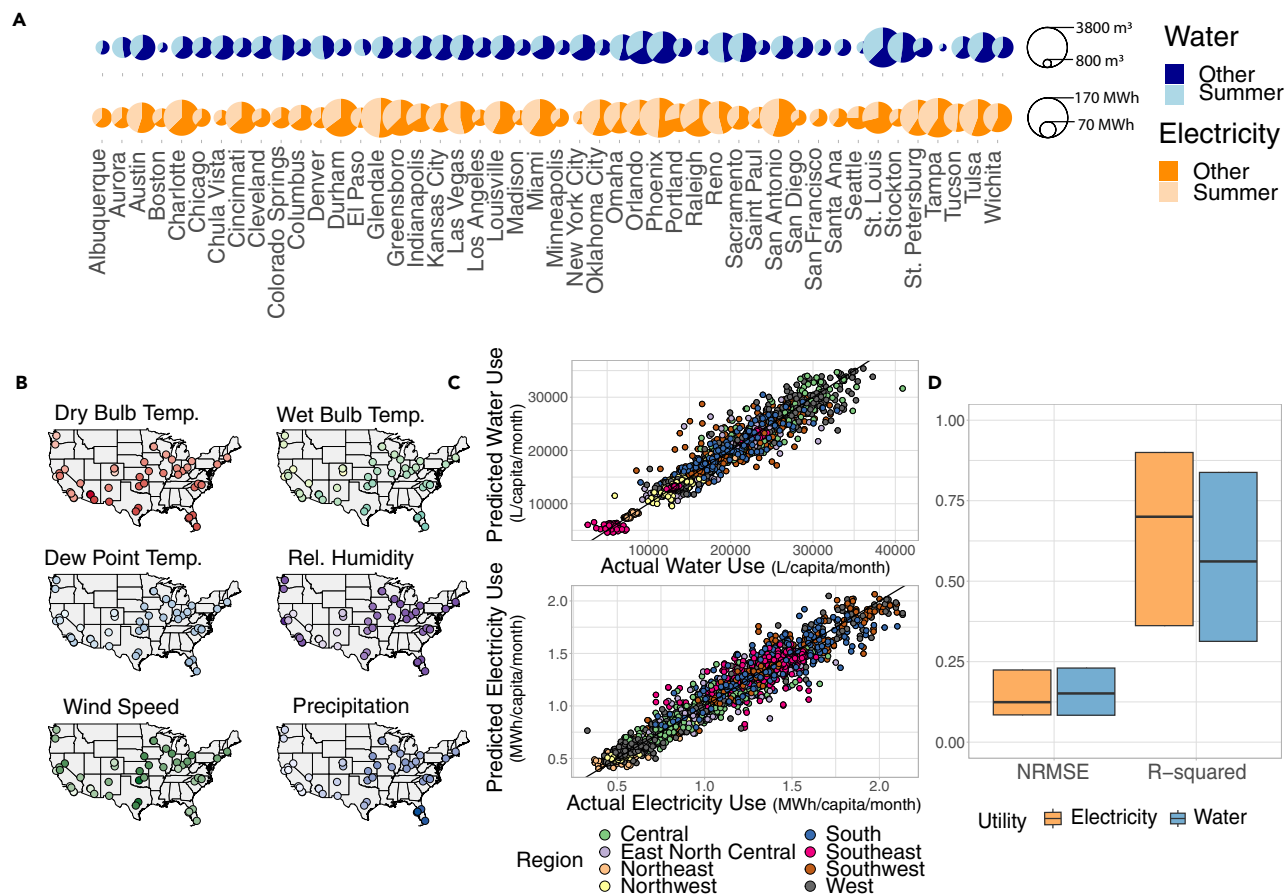


Figure 1. Data visualization and model performance

(A) Total water (m³) and electricity (MWh) consumption compared with the summer-only consumption for each city.

(B) Spatial distributions of the baseline input variables. We calculated several statistical measures (e.g., mean, maximum, minimum, sum, etc.) for each of these input variables, leading to a total of 17 predictors.

(C) The predicted climate-sensitive portion of the water and electricity use plotted against the actual data, with a 45° line for reference.

(D) Measure of error (normalized root-mean-square error [NRMSE] and R^2), plotted as a boxplot to represent the variance in model performance across the 46 models built (one model per city). The lower bound represents the minimum value, the upper bound represents the maximum value, and the crossbar represents the median.

has shown lower values of R^2 ,^{8,16} suggesting that it is not uncommon to see low R^2 values when modeling the climate-sensitive portion of the water-electricity nexus. Nonetheless, R^2 is not a measure of error and is therefore rarely used to evaluate *predictive* performance, which is usually characterized through measures such as RMSE (root mean square error) or MAE (mean absolute error). Here, we consider NRMSE the primary measure of predictive error.

Overall, the predictive model reliably characterized the historical records of the climate-sensitive portion of the coupled water and electricity demand, with median NRMSE values of 0.12 and 0.15 for electricity and water demand, respectively (Figure 1D). This suggests that the median error for water and electricity consumption was below 15%. In general, there was less variation in the NRMSE than the R^2 , indicating that the *predictive* performance of the model is relatively stable throughout the cities studied. This further indicates the transferability of the model, particularly when the goal is to predict the climate-sensitive portion of the water and electricity consumption. Going forward, this model

was used to project the coupled water and electricity demand under the high-emissions climate change scenario.

Regional trends in the projected demand

The primary results from the high warming projection analysis are shown in Figure 2 (see Figure S2 for moderate warming results). In general, the model is projecting increases in electricity demand (Figure 2B) across the country. Meanwhile, water demand (Figure 2C) is projected to increase slightly or remain the same across much of the US. The majority of these projected increases are occurring in the Midwestern cities, particularly with respect to electricity use. Previous work has shown significant increases in electricity use, as well as air conditioning use in particular, within the Midwest region.^{8,24} Our results align with these previous findings, indicating close to a 20% increase in the climate-sensitive portion of electricity demand by 2080, solely because of climate warming. Likewise, water consumption is expected to increase in many Midwestern cities. These projected increases are likely because of the analogs, which

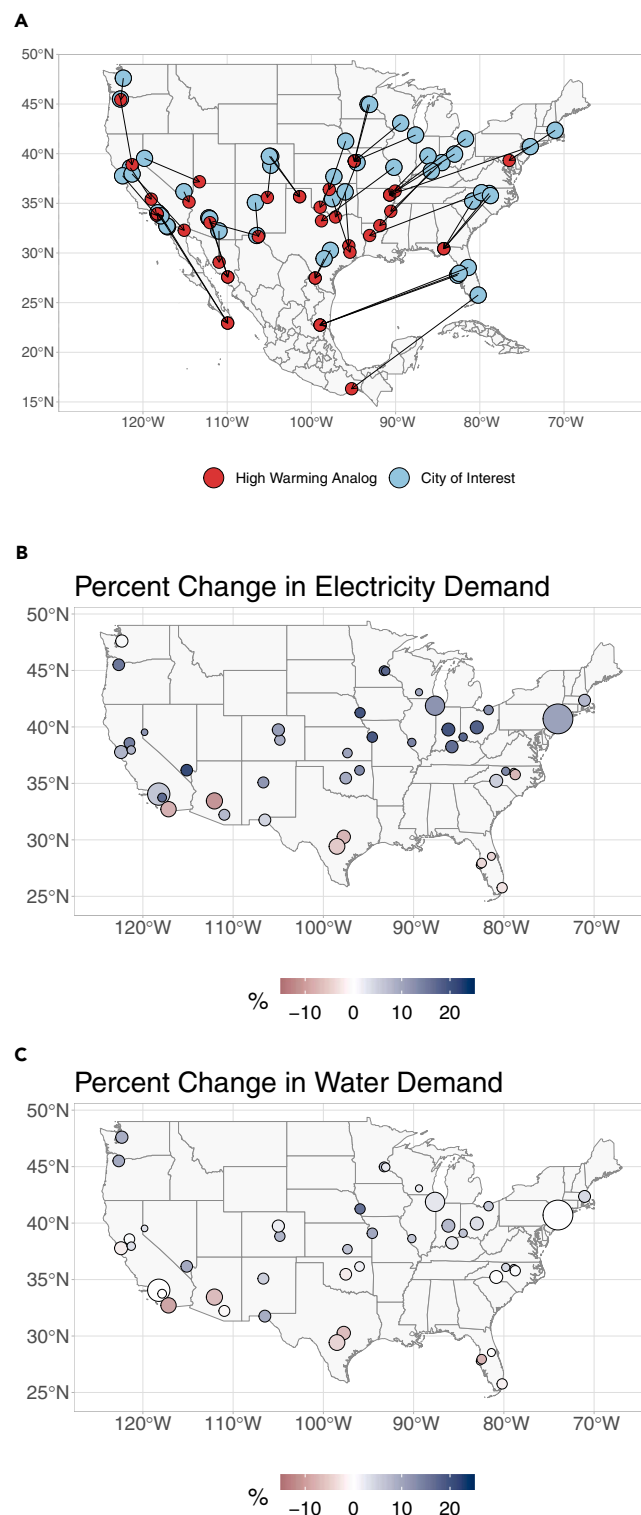


Figure 2. Climate analogs and model projections

(A) Target cities and their high warming analogs.
(B) Projected changes to electricity demand across the 46 cities (sized by population).
(C) Projected changes to the water demand across the 46 cities (sized by population).

are generally located in the southern US. This means that the average future under high warming for the Midwest will be warmer and drier, leading to increased demand for water and electricity within the residential sector. Previous work in the Midwest found similar results. In particular, Obringer et al.⁸ projected a 20% ($\pm 15\%$) increase in Chicago's electricity demand after surpassing the 2.0°C threshold, following the same water-energy nexus approach. The present analysis projected an increase of 12% for the city of Chicago's electricity use, which is within the estimated range reported previously. The primary difference between the two methodologies was the use of climate analogs rather than downscaled GCM data. The fact that the present results for Chicago are within the range of GCM-based projections provides greater confidence in the reliability of this method, at least for Chicago. Additional work is needed to confirm the results for the rest of the cities; however, given that this method is significantly less computationally expensive and more user friendly, it does signify the possibility of using climate analogs to evaluate climate change impacts.

In the Northeast, the electricity consumption is projected to increase under the high-emissions scenario, while water is projected to remain relatively constant or increase slightly. This indicates that the average high-emissions future has higher temperatures (which are often associated with higher electricity consumption in countries with high air conditioning penetration, such as the US) with limited changes to precipitation (which is often the primary climate driver of water consumption). In terms of water consumption, the mountain region and western US are projected to see increases. Given the ongoing issues of drought within the western United States,^{33–35} this increase could put additional stress on the water system. Further, the increased consumption of electricity paired with this increase in water consumption could negatively impact hydropower in the western US. Specifically, during periods of drought, the increased water consumption could create issues for the reservoirs, which may then limit the effectiveness of hydroelectric plant operations.^{9,10} Understanding these interconnections is critical for building resilience and planning for future climate change.

While most cities show an increase in water and electricity demand, there are some cities that are projected to see reductions in the climate-sensitive water and electricity demand, primarily those in the southern US, which tend to have analogs in more tropical regions (Figure 2A). These regional differences are exemplified in Figure 3. In general, the results indicate that, under the high-emissions scenario, the majority of the cities studied are projected to see increases in water and electricity demand up to 15% and 20% above current levels, respectively, although some cities are projected to see reductions in demand. These latter cases are found in the southern regions of the country, demonstrating a north-south gradient to projected changes in water and electricity demand, which will be further discussed in the next section.

Southern cities have a projected decrease in demand

Within the country as a whole, there is a gradient that separates the southern cities from the central and northern cities. In particular, we see reductions in water and electricity demand below this gradient, with increases above it. This gradient is perhaps counterintuitive because we would expect the warmer regions

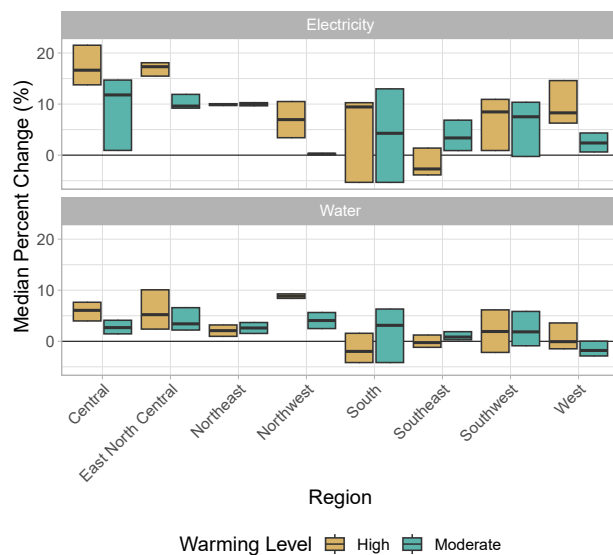


Figure 3. Regional projections of water and electricity demand

Shown are projected regional changes in water and electricity demand. The lower bound and upper bound represent the 25th and 75th percentiles, respectively, while the crossbar represents the median.

to get warmer (and potentially drier), which would lead to higher levels of electricity and water demand. However, by investigating the analogs of the southern cities evaluated in this analysis, we find that the analogs are often more tropical, leading to increased precipitation and slightly cooler (average) dry bulb temperatures. Based on the relationship between climate and the water-electricity demand nexus, this leads the model to project reductions in the *climate-sensitive* portion of water and electricity demand in these cities. It should be noted, however, that these results are based on the ensemble mean of 27 GCMs, thus there is some uncertainty in the initial climate analogs. Fitzpatrick and Dunn²⁵ discuss this uncertainty and point out that these counterintuitive results indicate that there are possible futures in which cities experience completely new climate zones.

For example, our results indicate that the city of Phoenix (Arizona) may experience significant reductions in residential water and electricity demand—approximately 15% in both sectors—because of the relationship between climate and the water-energy nexus without accounting for changes in population, socioeconomic status, cultural norms, or other anthropogenic impacts, such as urban heat islands. This is a potentially surprising result, particularly given the increasing levels of extreme heat that have occurred in the Phoenix area as well as the ongoing mega-drought, which requires further analysis.

To delve deeper into these counterintuitive results, we can investigate the important variables that are being used by the model to make the projections. When evaluating Phoenix, previous work found that the most likely climate analog, Esperanza, Mexico, is cooler and wetter in the summer compared with Phoenix. This may explain why our model, which only considers the climate impacts, is projecting reduced water and electricity demand under a high-emissions climate change scenario. Furthermore, our analysis found that the most important variable for predicting the climate-sensitive portion of the water-energy

nexus in Phoenix was the average dry bulb temperature (see Figures S1 and S5 for important variables by city and region, respectively), which was nearly 20% lower in Esperanza than Phoenix during the observation period. Given that higher average dry bulb temperatures tend to lead to higher electricity consumption within the observed values for the southwestern region (see Figure S3), it is likely that this reduction in temperature leads to a projection of reduced electricity demand because of the climatic conditions. Similarly, an increase in total precipitation would lead to a reduction in water consumption because of a reduction of the need for outdoor landscaping.

While the model itself is built on these relationships, it is important to note that the uncertainty in the future climate analogs could lead to different values. When we investigate the individual analogs for Phoenix, for example, we find that they range from Buckeye, Arizona (about 40 miles southwest of Phoenix) to several locations in the Sonora state of Mexico (including the cities of Nacozari de García, Hermosillo, and Esperanza). While the ensemble mean analog resulted in a projected 15% decrease in water and electricity consumption, the individual analogs lead to different results. In particular, if we consider Buckeye, Arizona the analog for Phoenix, the result is a 4.5% and 1.8% reduction in water and electricity consumption, respectively. Likewise, if we evaluate the other Mexican cities, the result is a 6.3% reduction and a 3.5% reduction in water consumption for Hermosillo and Nacozari de García, respectively. For these two analogs, the projected electricity consumption is 10.0% and 12.5% compared with the baseline. Thus, we can see that the different analogs lead to different magnitudes of change for the city of Phoenix, depending on the GCM used to derive the analog.

Likewise, cities in Florida are projected to see reductions in our modeling results. In particular, the city of Tampa is projected to see close to a 10% decrease in water demand with only a slight decrease in electricity. The climate analog of Tampa is Ciudad Mante, Mexico, which receives more precipitation in the summer than Tampa. This increase in total precipitation would likely lead to less outdoor water use, the main source of urban water demand during the summer months. Looking that the possible analogs for Tampa, they all fall along the Gulf coast in Mexico, ranging from Ciudad Mante (the ensemble mean analog) to Palenque in the state of Chiapas. Comparing Ciudad Mante with Palenque, we see that both have a tropical climate; however, the Köppen climate designation for Ciudad Mante is tropical savanna, while that of Palenque is tropical monsoon. These differences in climate are likely to cause differences in the projection of the climate-sensitive water and electricity demand in the city of Tampa. In fact, if we consider Palenque the analog for Tampa, then the resulting change is a 6.5% and 3.1% reduction in water and electricity consumption.

These counterintuitive results suggest a chance that residents will experience novel climatic conditions, at least for their region. Previous work has found that the chance of experiencing a novel climate increases as the future climate scenarios become more extreme (e.g., moving from RCP4.5 to RCP8.5).^{36,37} Further, the likelihood of localized novel climates has been shown to increase dramatically at low elevations,³⁷ which aligns with our results in the southern US. The authors demonstrated that higher elevations are more likely to have local analogs that are lower in elevation but otherwise maintain

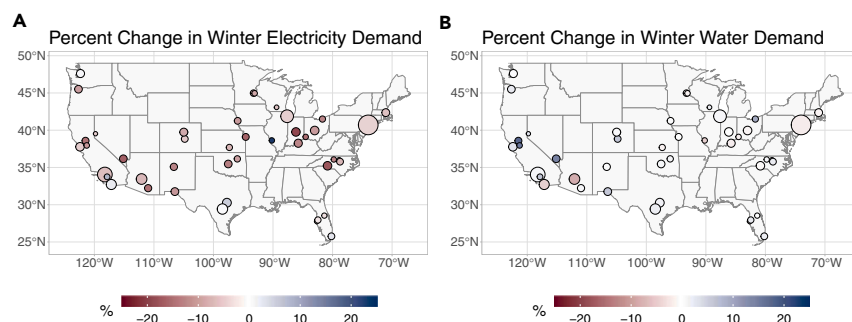


Figure 4. Projected demand shifts during the winter months

(A) Projected percent change for the electricity demand during the winter months under the high warming scenario.

(B) Projected percent change for the water demand during the winter months under the high warming scenario.

similar seasonal patterns. The lower elevation locations, however, are much more likely to have nonlocal analogs with vastly different seasonal patterns. In particular, Mahony et al.³⁷ found that locations such as the Sonoran desert (Arizona) and much of the Gulf Coast are likely to experience novel climates by 2100 under RCP8.5, aligning well with the north-south gradient shown in this study. These highly dissimilar analogs lead to novel climates in the southern US, which, in turn, leads to counterintuitive projections of climate-induced shifts in water and electricity demand in those areas.

Projected demand shifts during winter months

Throughout this study, we focused on the summer months; however, the remainder of the year is also critical for planning purposes. In Figure 4 we show the projected changes to electricity and water demand during the winter months under the high warming scenario (Figures S6 and S7 for results from the intermediate months and the moderate warming scenario). Compared with the summer months, there is no pronounced north-south gradient, with the majority of cities projected to experience reductions in winter electricity use and minimal changes to winter water use. Similar results for winter water and electricity consumption were found in the Midwest region.⁸ In particular, the decrease in winter electricity use is likely due to more milder winters leading to less need for electric space-heating, especially in northern cities. In south, there are a few cities with minimal reductions in electricity consumption (e.g., Tampa and Orlando), but there are other cities that are likely to see increases (e.g., Miami and Austin). These increases might be indicative of cities in which winter becomes more extreme (in terms of temperature), leading to increased use of electricity for space-cooling. Across the 46 cities studied here, the average reduction in winter electricity is 6.8% with a maximum of nearly 19%. This level of reduction, however, is not quite enough to offset the increase in summer electricity consumption across the US, where there was a mean increase of 7.7% and a maximum increase of 21.4% above baseline. Moreover, the larger swings in electricity consumption from season to season could create difficulties for grid planning and management, particularly in the central regions, which have some of the largest projected increases in summer electricity, followed by some of the largest decreases in winter electricity. In terms of water, while most cities show minimal changes (mean = 1.5% increase), cities in the west, including Las Vegas and several Californian cities, are projected to increase their winter water consumption, which could strain the water resources in the region.

Demand shifts under shared socioeconomic pathways

As shown in Figure 2B, the majority of the cities included in the analysis are projected to see increases in summer electricity demand, including the most populous cities in the country: New York City (New York), Chicago (Illinois), and Los Angeles (California). These cities show little to no change in the water demand, likely because of their dense nature, which does not allow much outdoor use, but with large increases in summer electricity use (6%–12%) per capita. However, it is unlikely that the population of these cities will remain as it currently is, so even relatively small increases in per-capita demand could have a large impact on the total demand in the future. To understand the impact of population growth on the total demand for services, we evaluate the increase in electricity use based on the shared socioeconomic pathways (SSPs) of the major cities mentioned above, the results of which are shown in Figure 5.

In terms of the total change in electricity demand, a 6%–12% increase in per-capita electricity demand in these three major cities represents an increase of 2.5–9 million MWh in a given summer month, depending on the city and population growth scenario (Figure 5). In Figure 5, we show the total monthly electricity increases in the counties where Chicago, Los Angeles, and New York City reside, based on the five SSPs.³⁸ The SSP data are aggregated by county rather than city, so the total electricity values are for the counties as a whole (Cook County for Chicago; the combination of Bronx County, King County, New York County, Queens County, and Richmond County for New York City; and Los Angeles County for Los Angeles). We make the assumption that the residents beyond the city limits will have the same increase in electricity demand, given the similarity in climatic changes that will be experienced. Within these pathways, SSP 1 represents sustainable development, while SSP 5 represents fossil-fueled development.³⁸ Because SSP 5 is more in line with the worst-case warming scenario, we can use the total electricity from that scenario to better understand the possible changes to the electricity demand patterns under extreme climate change. In particular, the results presented in Figure 5 show that, in 2080, Chicago will require 5.7 million MWh in a given summer month compared with the 4.3 million MWh generated now. Likewise, Los Angeles will require 9.2 million MWh (compared with 5.4 million MWh today), and New York City will need to provide 9.0 million MWh compared with the 4.2 million MWh required currently. These increases could create additional expenses for the utilities, particularly if they aim to increase the share of renewable energy sources to help mitigate climate change.

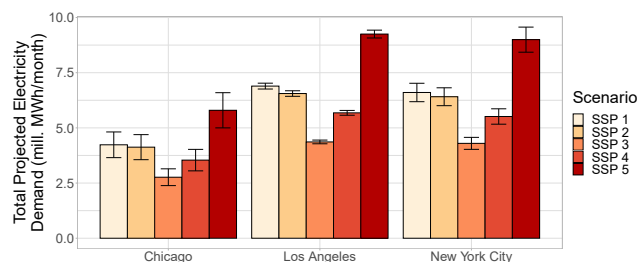


Figure 5. Demand shifts under SSPs

Shown is the total projected electricity demand increase by 2080 in three major cities under the five shared socioeconomic pathways (SSPs).³⁸ Data are represented as the mean $\pm$ standard deviation.

DISCUSSION

In this study, we presented a novel approach to climate change impact analysis, using climate analogs as a proxy for traditional GCM-based future climate data. We used these climate analogs to project the changes to water and electricity demand across 46 US cities under the worst-case climate scenario. In particular, the projection analysis demonstrated that some cities may experience up to 15%–20% more demand for water and electricity because of climate change, while others may see reductions, depending on their analog. These results have implications beyond the modeling itself. For example, our analysis showed a possible increase of 3.8 million MWh of electricity across Los Angeles County in the worst-case climate scenario (RCP8.5 + SSP 5). If this climate-induced change in demand is not accounted for, then it could become more difficult for cities to reach renewable energy targets. For example, Los Angeles plans to achieve 100% renewable energy by 2050.³⁹ An increase of 3.8 million MWh across the county would require 14,000 additional 1.5-MW wind turbines, approximately 20% of the current operational stock in the US, just for the relatively small area servicing Los Angeles County. If cities are not prepared for these changes, then they may fall back on carbon-intensive electricity generation, which will ultimately exacerbate the climate crisis.

In terms of the impacts on water and electricity consumption, the results presented here represent the average increase over the course of a single summer month. There will be, however, extreme events, such as droughts or heatwaves, that lead to peaks in water and electricity consumption beyond our results. These events put a strain on the infrastructure systems, with managers and operators often resorting to conservation mandates^{40,41} or, in the case of the electric grid, rolling blackouts.^{42,43} Previous research has shown that vulnerable groups, such as lower-income families, the elderly, or racially/ethnically marginalized people, experience more extreme impacts during heatwaves,⁴⁴ leading to public health crises. Should utilities and management agencies fail to account for the climate-induced shifts in demand, the resulting supply deficits should result in increased vulnerability to heat-related issues. To mitigate these potential impacts, it is critical to build resilient infrastructure that can handle the climate-induced demand increases. This ultimately comes down to working with decision-makers and infrastructure managers.

Decision-making implications of the analysis

There are a number of decision-making implications of this work, primarily related to the management of utilities and the planning of future infrastructure projects related to water and electricity. Decisions for upgrading or expanding water supply or electricity generation infrastructure often depend on a number of factors, from projected population increases to the impact of climate change. However, many of these decisions are made by individual utilities without accounting for the interconnection between water and electricity demand.¹⁴ The results presented in this study further expand the use of multi-outcome modeling to evaluate possible changes in demand for water and electricity consumption simultaneously, which has been shown to be a more accurate approach for both systems.¹⁶ Prior to this study, however, there was not a large-scale analysis that could be used across the US to demonstrate the value of these critical interconnections. Additionally, many utilities are aware of the possible climate change impacts and are attempting to take measures to build resilience and adaptive capacity. However, there remains a significant computational gap between what is needed to run high-fidelity, downscaled GCMs and the tools and skillset available to many local utilities. Therefore, there is a need to develop alternative means of analysis to provide means for these smaller, local utilities to understand the impact of climate change on their systems. The results from this work demonstrate that climate analogs are a viable means for conducting a climate change impact assessment for water-electricity demand. Given that there are 540 North American cities included in the initial analog assessment,²⁵ there is ample opportunity for smaller municipalities to find their most likely analog and then use the model and results presented here to evaluate possible climate-induced changes to their demand structure.

Furthermore, the results in this work highlight several critical challenges that water and electricity utilities will face under climate change. First, the model is projecting significant increases in summer electricity consumption across many US cities, particularly in the northern parts of the country. Planning for these climate-induced demand shifts will need to occur simultaneously with planning for shifting socioeconomic conditions, population growth, and demographic changes, thus adding an extra layer of complexity. Should utilities not prepare for these increases, it is possible that there will be supply deficiencies, possibly resulting in blackouts during critical periods, such as heatwaves. Further, the existence of the summertime north-south gradient could lead to difficulties maintaining adequate supply across the interconnected grids. The need to transfer electricity from the south to the north would increase expenses for customers and potentially create logistical issues for the managing bodies. Finally, the difference in summer and winter electricity could create issues for managing the grid. In particular, the extreme swings between lower demand and higher demand can create issues for balancing the electricity across the grid. For example, a reduction in winter electricity could mean that a number of powerplants are not being used but still need to be maintained and kept online for support during the increases projected for the summer months. This is not an uncommon practice in the current day, particularly because many residences rely on natural gas for heating purposes. However, the projection of more extreme swings between winter and summer

could exacerbate the existing issues, particularly if utilities and management agencies are not prepared for these climate-induced changes.

When decision-makers understand how climate change might impact demand, they can enact policies to prepare for that increase, such as expanding supply or generation capacity or requiring more efficient appliances in buildings. Moreover, the multi-outcome model used in this analysis, multivariate tree boosting, is generally more interpretable than other “black box” machine learning algorithms. This means that users of the model can extract insights and understand the nature of the relationships being used to make the projections. That being said, while the nature of the model is relatively straightforward and interpretable, there is still a need to improve the uncertainty quantification.

Sources of uncertainty in the projection analysis

There are several areas of uncertainty within this study. For example, the projection analysis conducted in this study (and shown in [Figures 2 and 3](#)) uses the climate analogs developed by Fitzpatrick and Dunn²⁵ through an ensemble mean of 27 GCMs. Because of the differences in the GCMs used in the initial climate analog analysis, there is a level of uncertainty that is carried over into the present analysis. In particular, there are 28 possible analogs for each city (one per GCM plus the ensemble mean). The uncertainty associated with the 28 possible analogs could lead to some differences in projections, as discussed above for the cities of Phoenix and Tampa. To further evaluate this uncertainty among different analogs, we randomly selected eight cities (one per region) and projected the change in summertime residential water-electricity demand using a range of 2–4 individual analogs, depending on the city, plus the ensemble mean analog. The extra analogs were selected to show the range of analogs based on the 27 GCMs leveraged by Fitzpatrick and Dunn.²⁵ These individual analogs are shown in [Figure S8A](#). The results from this uncertainty analysis are shown in [Figure S8B](#). In general, the difference in projected changes across different analogs was found to be within reasonable bounds. For example, the majority of the cities tested did not show a change in sign (e.g., an ensemble mean projected increase with an individual analog projected decrease) with relatively small ranges, particularly with water consumption. An exception to this pattern is the city of Raleigh. When we used the ensemble mean analog for Raleigh (Tallahassee, Florida) the model projected a reduction in electricity consumption. However, the range of individual analogs for Raleigh includes Sabinas, Coahuila (Mexico) and Waco, Texas, both of which lead to projected increases in the electricity demand. This uncertainty speaks to the possibility of novel climates, as has been shown to be highly probable in the southeast US under RCP8.5.³⁷ That being said, the majority of cities had more agreement among the different analogs, with several ensemble mean analogs occurring close to the median projected changes. Therefore, this allows us to evaluate the ability of using the ensemble mean analog as a proxy for traditional GCM-based impact assessments across US cities. It suggests that, in the cities and regions where the ensemble mean analog is in a completely different climatic zone, extreme climate change could lead to unexpected changes, creating further issues with quantifying and communicating uncertainties. As the

GCMs continue to evolve and improve, it is likely that the range of possible climate analogs will be reduced, leading to more certain results.

Another area of uncertainty lies within the electricity and water consumption data used to train the model. In this study, we collected residential usage data from local utilities and government agencies, but there is always a chance of an error within the data collection phase. For example, many local water utilities provided us with data obtained from their own data records (e.g., how much water was sold to residential customers in a given month), which we normalized over the service population. However, if a utility entered an erroneous value into their database, then it could skew the normalization and lead to uncertain results. To avoid this possible issue, we analyzed the data for outliers and did not find any. Nonetheless, there remains some uncertainty with regard to the data itself. Finally, there are issues relating to epistemic uncertainty. In this analysis, epistemic uncertainty refers to the uncertainty in the model assumptions. The quantification of epistemic uncertainty is an ongoing challenge within the data-driven modeling community.⁴⁵ In this analysis, we used a non-parametric modeling algorithm, which learns the relationship between the response and predictor variables, so there are no assumptions about linearity, normality, etc. However, when we conducted the projection analysis, we made the assumption that this learned relationship was stationary. That is, that the learned relationship between water-electricity demand and the climatic conditions would remain the same into the future. This relationship, however, may change in the future because of a number of factors (e.g., technology, cultural norms, or demographics) that were not included in this analysis. In this sense, there is a level of uncertainty that is not quantified in this study. Recent studies suggest that using multiple models or Bayesian processes can help quantify this epistemic uncertainty;^{45,46} however, such an analysis is beyond the scope of this study.

Study limitations and opportunities for next steps

This study focused on investigating the climate-induced changes to the water-electricity demand nexus across the US. In this sense, we only evaluated the impact of climate change on the demand structure. However, there are a number of other areas that can impact the demand, including culture, socioeconomic variables, demographics, and building characteristics. In particular, previous research has shown that increasing the efficiency of air conditioners can maintain the “status quo” in regard to total electricity consumption, even if people are using air conditioning more often because of climate change.⁴⁷ Similar work in the water sector has demonstrated how social norms can lead to reductions in water consumption beyond what modelers may expect based on the climate and housing characteristics.⁴⁸ While including these critical areas of influence was beyond the scope of this study, future research can expand upon the climate analysis to include these key features.

Within this analysis, we leveraged a methodology that focused on projecting the average changes to the water-electricity demand nexus for an average month under climate change scenarios. Therefore, the analysis is not able to evaluate specific extreme events, such as heatwaves or droughts, or inter-annual

variability. Moreover, the analysis accounts for the monthly per-capita demand across a given city rather than peak demand. This masks both high- and low-intensity users in favor of evaluating the average. There has been recent work to suggest that the climate response within the electricity sector is asymmetrical with regard to rising temperatures.⁴⁹ Therefore, separating out the high-intensity users can be beneficial for improving the overall predictive accuracy across a city. While this type of analysis was beyond the scope of this work, which aimed to present relative projections for the whole city under long-term climate change, future research can leverage high-resolution datasets to evaluate the differences between high- and low- or moderate-intensity users under climate change as well as extreme events.

Finally, the climate analogs were initially developed using the CMIP5 suite of GCMs.²⁵ As the CMIP6 models become more commonplace and downscaled, bias-corrected data become more readily available from these next-generation GCMs, and it might be beneficial to develop new climate analogs. Additionally, in the original study, the authors only considered maximum temperature, minimum temperature, and precipitation to determine the most likely analog. However, recent work has indicated that relative humidity and other measures of temperature play a major role in climate change impacts;²³ therefore, including these climatic variables might improve the climate analog development, particularly for use in impact assessments. While the aim of this study was not to criticize or improve upon the climate analogs, making these improvements would like help to build up the use of climate analogs as a modeling and communication tool as well as provide greater certainty around future projections in the climate change impact space.

Conclusion

In this study, we build a scalable and transferable model for projecting the coupled water and electricity demand into the future under climate change. In particular, we leverage climate analogs as a modeling tool to make future projections across 46 cities in the US under a high-emissions climate change scenario. Moreover, we use a water-energy nexus framing, which models the two sectors simultaneously and has been shown to be more representative of the real-world systems.^{14,16} The results indicate increases in summertime climate-sensitive water and electricity demand up to 15% and 20%, respectively. In addition to the worst-case scenario, we also evaluated the changes under a moderate warming scenario (RCP4.5; Figure S2). The results demonstrate a significant improvement across much of the United States under less intense warming. This is particularly true for electricity demand in the central, east north central, northwest, and west regions, which is in line with previous work. However, there was little change in the northeast, south, and southwest regions for water and electricity demand. This suggests that the analogs for both warming scenarios are similar across these regions, which leads to relatively similar climates regardless of the intensity of climate change. Overall, these differences between the two scenarios provide further support for maintaining a lower warming intensity in terms of limiting increases in water and electricity demand across the US. In the absence of proactive mitigation strategies, the unanticipated in-

creases in demand could lead to supply shortages and complicate the transition to clean energy. Moreover, previous research has indicated that vulnerable groups, such as low-income families or the elderly, are likely to experience more intense impacts in the event of supply shortages. Therefore, preparing for the climate-induced shifts in water and electricity demand is not only an issue of climate change adaptation but also one of environmental justice.

Going forward, it is important to continue to develop future projections of water and electricity demand that not only account for the interdependencies between the two sectors but also the impact that climate change will have on the coupled demand structure. Ultimately, this coupled demand will feed back into the supply side, which is also deeply interconnected (e.g., water is necessary for many modes of electricity generation, so higher demand for electricity not only leads to more water demand at the end use but also at the generator). In this sense, by making these interconnected projections, we can also encourage interconnected action, such as the electricity utilities working to encourage water conservation as a way to maintain adequate reservoir levels for hydropower generation. This is also critical for the shift to renewable energy because planning electricity generation requires accurate representations of the demand structure. For example, energy system plans that do not account for the climate impact on demand or that this impact is likely to affect the water sector, which, in turn, will affect the electricity sector, may not achieve the necessary capacity to provide adequate supply for future generations. In this sense, the modeling of climate impacts through a water-energy nexus framing is critical for furthering climate change mitigation and adaptation efforts. Moreover, the use of climate analogs as a proxy for future climate data can greatly reduce the computational needs, potentially providing practitioners with a more interpretable way to model and communicate the impacts of climate change on their local systems. Finally, although this work focused on the climate-sensitive portion of water and electricity demand, the methods can be easily integrated with work investigating the impact of technological change and population growth, through SSPs, for example, as demonstrated for three major cities above. Additionally, the model can be scaled to encompass regions or transferred to other cities, states, or countries, provided the data are available. This type of interdisciplinary work is critical for building resilience to climate change and finding equitable solutions that are accessible to local governments and organizations.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Renee Obringer (obringer@psu.edu).

Materials availability

The study did not generate new unique materials.

Data and code availability

All data and code for this study have been deposited at Zenodo under the <https://doi.org/10.5281/zenodo.7826348> and are publicly available as of the date of publication. Additionally, the data and code are available via GitHub (https://github.com/reneeobringer/ClimateAnalog_WEN). Any additional

information required to reanalyze the data or rerun the models is available from the lead contact upon request.

Data description

In this study there were two types of data: consumption data and climate data. The consumption data included residential water and electricity consumption, which were obtained from different sources. The water consumption data were collected directly from the water utilities in the selected cities via Freedom of Information Act (FOIA) requests. Initially, cities with populations greater than 250,000 were selected, resulting in 84 initial cities. Of these 84 cities, only 46 cities (see Table S1 or Figure 1A for the list of included cities) responded with adequate data for our analysis. In particular, some cities simply did not respond the request, while others were unable to provide monthly water consumption data for the time period of interest: 2007–2018. The electricity consumption data were obtained from the US Energy Information Administration (EIA), which keeps values of monthly electricity use for a number of utilities across the US.⁵⁰ The use of a monthly timescale was driven by the data availability for water and electricity. The EIA data, for example, are collected and published at monthly and annual scales, while most local water utilities were likewise only able to provide monthly totals. Therefore, we opted to use a monthly timescale in this analysis. Finally, water consumption and electricity consumption were normalized by service population. The normalization process was important to allow a closer comparison between cities as well as the water and electricity consumption within a city because many water and electric utilities have different jurisdictions. Additionally, the trend was adjusted to remove the impact of non-climatic factors (i.e., technology, socioeconomic status) following a method by Sailor and Muñoz.⁵¹ In this method, the user calculates a yearly adjustment factor and applies it to each month in the dataset. This process allowed us to focus on the climate-sensitive portion of the water and electricity consumption rather than including the technological and socioeconomic factors that also influence the demand profiles. The trend adjustment method has been successfully used in a number of energy- and water-related studies (e.g., Alipour et al.,⁵² Obringer et al.¹⁶ and Obringer et al.³²). More information on the trend adjustment process can be found in the Supplemental experimental procedures.

The climate data were collected from the North American Regional Reanalysis (NARR).⁵³ The NARR data are available at a daily timescale and 0.3° (32 km) spatial resolution. The variables included dry bulb temperature, wet bulb temperature, dew point temperature, relative humidity, wind speed, and precipitation. Initially, the data were collected as daily values for the entire time period (2007–2018) in each city but were later aggregated to monthly variables to match the water and electricity demand data. In particular, the temperature, humidity, and wind speed variables were aggregated to form minimum, maximum, and mean monthly values, while the precipitation was aggregated to be the accumulated monthly total as well as the number of days within the month in which precipitation was recorded. Previous work has leveraged the NARR dataset for observational climate data, following a similar collection process as in this study.²³ Because we used the climate analog approach, there was no need to run any simulations to obtain the future climate variables. Rather, the climate variables from the various analog cities were used to represent the future climate of the cities of interest.

The climate analogs published by Fitzpatrick and Dunn²⁵ were developed using minimum and maximum temperature and total precipitation for the four climatological seasons. These analogs were calculated based on the sigma dissimilarity between these 12 variables (3 measures $\times$ 4 seasons) in the present-day city's and future city's climate. Generally, the authors aimed to find the location with less than 2σ difference between the cities, although some cities did have differences up to 8σ , depending on the GCM. Ultimately, each present-day city has 27 possible analogs using each of the selected GCMs plus an ensemble mean analog. For example, New York City's ensemble mean climate analog for the year 2080 (RCP8.5) is Jonesboro, Arkansas. This means that, in 2080, New York City can expect a climate similar to present-day Jonesboro if there is no global reduction in emissions. Therefore, to obtain future climate data for New York City, we collected observed climate data from Jonesboro (2007–2018) using the NARR dataset. In this study, we used the ensemble analog climate data as a proxy for a traditional GCM-based impact assessment. See the Supplemental experimental procedures for more information on the development of the climate analogs.

Overview of the methods and use of climate analogs

As discussed above, this study leveraged climate analogs developed by Fitzpatrick and Dunn.²⁵ In particular, we used the ensemble mean analogs to estimate the future water and electricity demand in the selected cities. This process covered six key steps, which will be discussed below. Further details can be found in the Supplemental experimental procedures and Figure S4.

First, we extracted the climate analog information. This involved identifying the analogs for each of our target cities through the online tool developed by Fitzpatrick and Dunn (<https://fitzlab.shinyapps.io/cityapp/>). After we identified the moderate and high warming analogs, we extracted the latitude and longitude for each analog and target city. The second step was to collect the climate data from NARR. For this step, we used the previously collected latitude and longitude data to find the nearest grid cell for each analog and target city. These data were collected at the daily scale. Therefore, the third step was to aggregate the data into monthly minimums, means, maximums, and totals. Fourth, we combined the de-trended consumption data and aggregated climate data into data frames for the modeling analysis. Finally, we conducted the modeling analysis and calculated the percent change, both of which are discussed below.

Modeling framework and analysis

The modeling framework was adapted from Obringer et al.,⁸ who projected the water and electricity demand into the future for six cities in the Midwestern US. The framework is centered around a statistical learning algorithm, known as multivariate tree boosting,³¹ which is capable of simultaneously predicting multiple variables. See the Supplemental experimental procedures for an algorithmic representation of this method. The algorithm also accounts for the interdependency between the response variables to improve the predictive accuracy, making the water-electricity nexus an ideal study application for the model. In this study, there are six key steps in the modeling process, as discussed above. For a more detailed description, see the Supplemental experimental procedures and Figure S4. In terms of the modeling framework, we first conducted a variable selection process through backward elimination. Next, we implemented the model through randomized cross-validation with the current data. Then, we conducted a projection analysis with the analog data. Finally, we evaluated the change in water and electricity consumption between the predictions (i.e., the current-day model) and the projections (i.e., the future model using climate analogs). Each of these steps is discussed below.

This work expanded the methodology of Obringer et al.⁸ to the entire continental US as well as changed several procedures within the framework. Namely, the model was first developed at a regional scale, grouping the cities based on the National Oceanic and Atmospheric Administration (NOAA) climate zones.⁵⁴ Within this step, we performed variable selection to reduce the number of variables within each region. Given that climatic patterns often evolve at the region level, this step was important for determining which variables were important throughout a region before delving into the city-level analysis. This regional analysis resulted in 4–6 important variables per region rather than the original 17, reducing the complexity and improving model accuracy. See Figures S1 and S5 for the important variables selected for each region. Following the region-level variable selection, we performed individual city analyses with the previously selected variables. First, to avoid overfitting, the model was trained and rigorously tested via randomized cross-validation using observational climate data collected from NARR and water and electricity consumption data from local utilities. This randomization breaks up the time series, removing any autocorrelation present. Therefore, we are able to model the relationship between climate and coupled water-electricity demand for any given month. Then, using this trained model, we substituted the observed climate data with NARR climate data from the analogs, creating projections of water and electricity demand. Finally, we evaluated the percent change between the analog-based projection and observed model results for each city. For more information on the methods used in this study, see the Supplemental experimental procedures as well as Figure S4.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.oneear.2023.10.004>.

ACKNOWLEDGMENTS

The authors would like to acknowledge support from NSF grants 1826161 and 1832688, as well as the National Socio-Environmental Synthesis Center (SESYNC) under funding received from NSF grant DBI-1639145. The authors would also like to acknowledge support from the Graduate School at Purdue University, the Bilsland Dissertation Fellowship, the Purdue University Center for the Environment, and the Purdue Climate Change Research Center. The authors are grateful to Matthew C. Fitzpatrick and Robert R. Dunn, who made their climate analogs publicly available online (<https://fitzlab.shinyapps.io/cityapp/>), which served as the basis for this study. Finally, the authors would like to thank Megan Rodder for help in the initial data collection phase.

AUTHOR CONTRIBUTIONS

Conceptualization, R.O., R.N., and R.K.; data curation, R.O., J.K., and R.K.; formal analysis and investigation, R.O.; writing – original draft, R.O.; writing – review and editing, R.O., J.K., R.N., R.K., and K.M.; visualization, R.O. and R.K.; supervision, R.N., R.K., and K.M.; funding acquisition, R.N.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

Received: February 10, 2023

Revised: July 14, 2023

Accepted: October 3, 2023

Published: October 31, 2023

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